CROSS-REGIONAL TRANSFERABILITY OF EUROSAT-PRETRAINED RESNET-50 FOR LAND USE AND LAND COVER CLASSIFICATION IN SOUTH ASIA:

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Abstract

Using deep learning for land use and land cover (LULC) classification has greatly progressed with benchmark datasets like EuroSAT, which contains 27,000 labeled Sentinel-2 satellite image patches covering 10 land cover classes within European regions. Despite this, applying models trained on EuroSAT to areas that differ geographically, climatically, and spectrally such as rapidly urbanizing South Asian cities remains a significant and underexplored research challenge. This study systematically examines how well ResNet-50 can transfer across regions under three experimental setups applied to Lahore, Pakistan: (1) a Baseline ResNet-50 pretrained on ImageNet and fine-tuned with Lahore Sentinel-2 data; (2) Model A a EuroSAT-pretrained ResNet-50 tested directly on Lahore without fine-tuning; and (3) Model B a EuroSAT-pretrained ResNet-50 fine-tuned with Lahore-specific labels. Lahore is selected because it is among South Asia's fastest-growing urban areas, with over 241% urban expansion recently. Using metrics like Overall Accuracy (OA), class-wise F1-score, Cohen's Kappa (κ), and t-SNE visualizations of features, the study measures the domain differences between European EuroSAT and South Asian Lahore Sentinel-2 images and evaluates the benefits of domain-specific pretraining versus general ImageNet pretraining. The results include a validated, reproducible deep learning pipeline for LULC classification, a publicly accessible labeled Sentinel-2 dataset for Lahore, and practical sample size guidelines for South Asian regions practitioners.

Keywords: EuroSAT, ResNet-50, Transfer Learning, Domain Shift, Land Use Land Cover, Sentinel-2, Lahore, Deep Learning, Remote Sensing, Domain Adaptation

1 Introduction

Accurate and quick mapping of land use and land cover (LULC) is crucial for urban planning, environmental monitoring, agriculture, climate change assessment, and SDG tracking. Remote sensing, particularly multispectral satellite imagery from ESA' s Sentinel- 2, provides high- resolution, freely accessible spatial data for LULC analysis at regional and global scales. Combining Sentinel- 2 images with deep learning has greatly improved classification accuracy, surpassing traditional pixel or machine learning methods, thus transforming LULC mapping. However, the applicability of deep learning models trained on European datasets to very different landscapes such as the diverse, semi- arid urban areas of South Asia is still uncertain and underexplored.

EuroSAT, developed by Helber et al. (2019), is the most common benchmark dataset for LULC classification using Sentinel- 2 imagery. It contains 27, 000 geo- referenced image patches, each 64 × 64 pixels at 10- meter resolution, covering 10 land cover types across Europe. The dataset has two versions: an RGB version (3 bands) and a multispectral version (13 Sentinel- 2 bands). Deep learning models trained on EuroSAT, especially CNNs and Vision Transformers, reach over 98% accuracy, establishing it as a reliable performance benchmark. Nonetheless, EuroSAT captures only European environments temperate climates, structured farms, dense vegetation which are quite different from the arid, semi- arid, and diverse urban landscapes of South Asia.

Lahore, the Punjab provincial capital and Pakistan' s second- largest city, exemplifies the rapid urban transformation seen across South Asia. Its built- up area expanded from around 470 km ^{2} in 2003 to 956 km ^{2} in 2023, a 28% increase in twenty years. Today, settlements occupy 41. 14% of Lahore District (68, 68,015.98. 98 hectares), while agricultural land declined by over 31% between 1993 and 2023. This swift change results from population growth, migration, and unplanned urban sprawl, creating a complex, heterogeneous LULC landscape that needs accurate, scalable, and cost- effective monitoring. Despite this pressing need, there are no high- quality labeled deep learning datasets for Lahore' s Sentinel- 2 images in existing research.

Applying a model trained on EuroSAT' s European data to Lahore presents a major domain shift: spectral values, spatial textures, and class boundaries vary significantly. Lahore' s dense informal settlements look different spectrally from European residential areas; irrigated mango orchards differ from European crops; dusty barren land has no direct equivalent in EuroSAT. Quantifying this domain shift, understanding whether EuroSAT pretraining outperforms ImageNet pretraining for South Asian remote sensing, and determining the minimum number of locally labeled samples needed for effective fine- tuning are vital scientific and practical questions that motivate this research..

2 Problem Statement

Existing deep learning studies on EuroSAT LULC classification have only been validated in European environments and haven't assessed their effectiveness in South Asian landscapes. The domain shift caused by differences in spectral data, climate, and land-use patterns between Europe and South Asia, particularly Lahore, Pakistan, remains unquantified. Furthermore, no research has systematically compared the benefits of EuroSAT-specific pretraining versus ImageNet pretraining for cross-regional satellite image classification within a controlled, architecture-fixed experimental framework. This knowledge gap hampers practitioners in data-scarce South Asian regions from making well-informed decisions about the best pretraining and fine-tuning strategies for their LULC tasks. The absence of a publicly accessible, labeled Sentinel-2 LULC dataset for major Pakistani or South Asian cities complicates reproducible deep learning research in the region.

3 Research Objectives

This study is guided by seven clear and measurable objectives. The first, Baseline Training, involves training and evaluating a ResNet-50 model with ImageNet pretraining, then fine-tuning it on Lahore Sentinel-2 LULC data to establish a performance benchmark for comparison with future models. The second, Zero-Shot Evaluation, tests how well a EuroSAT-pretrained ResNet-50 (Model A) classifies Lahore's land cover without domain-specific fine-tuning, providing a direct measure of cross-regional transferability from Europe to South Asia. The third, Domain-Adapted Fine-Tuning, further fine-tunes an EuroSAT-pretrained ResNet-50 (Model B) on Lahore data to see if local adaptation improves performance beyond the ImageNet baseline. The fourth, Domain Shift Quantification, investigates performance differences by measuring spectral and structural differences between European and South Asian Sentinel-2 images through t-SNE visualizations and class accuracy comparisons. The last three objectives focus on practical and open-science goals: The fifth, Sample Efficiency Analysis, finds the minimum number of local samples needed per class for Model B to reach over 85% Overall Accuracy, tested at sample sizes of 50, 100, 200, 300, and 500. The sixth, Class-Level Transfer Analysis, examines which land cover classes transfer most effectively from EuroSAT to Lahore, using per-class ΔF1 domain gap scores and explaining patterns with spectral and landscape factors. Finally, the seventh, Open-Science Contribution, commits to releasing the Lahore Sentinel-2 LULC dataset (2023) under Creative Commons CC-BY 4.0 on Zenodo, along with all model weights and code on GitHub, ensuring reproducibility and community access.

4 Research Questions

- To what extent does an EuroSAT-pretrained ResNet-50 (Model A) accurately classify land use and land cover in Lahore, Pakistan, under a zero-shot transfer setting, and does domain-specific remote sensing pretraining on EuroSAT provide a statistically significant classification advantage over general-purpose ImageNet initialisation in the absence of target-domain fine-tuning?
- To what extent does full-network fine-tuning of an EuroSAT-pretrained ResNet-50 (Model B) on Lahore-specific labelled samples lead to a significant improvement in overall accuracy compared to an ImageNet-pretrained baseline fine-tuned under identical hyperparameter conditions, and what is the magnitude of this performance difference in percentage points?
- What is the minimum number of locally acquired, class-balanced labelled training samples per land cover category needed for Model B to achieve an acceptable classification performance defined as an Overall Accuracy exceeding 85% when fine-tuned from an EuroSAT-pretrained ResNet-50 checkpoint on Lahore Sentinel-2 imagery?
- Which land use and land cover classes show the most significant domain shift between the European EuroSAT source domain and the South Asian Lahore target domain, as indicated by per-class ΔF1 scores? Additionally, what spectral reflectance properties, spatial structural features, or landscape-level factors explain the observed cross-regional transferability patterns?

5 Literature Review

Since its debut, EuroSAT has become the standard benchmark for LULC classification using Sentinel-2 images. Helber et al. (2019) showed that ResNet-based CNNs reach 98.57% accuracy on this 10-class benchmark, setting a high performance standard. More recent research has built on this, with EfficientNet-B0 attaining 98.1% accuracy and Vision Transformers achieving state-of-the-art results with multi-year temporal data (Wang et al., 2024).

However, all available EuroSAT benchmark evaluations focus solely on the European EuroSAT test split, neglecting the aspect of geographic generalization a gap this research directly tackles. To date, no published studies have tested EuroSAT-pretrained models on any South Asian target domain, making this investigation the first of its kind. ResNet-50 (He et al., 2016), a 50-layer deep residual network, incorporates skip connections to combat the vanishing gradient problem, allowing for the efficient training of very deep networks. In remote sensing, ResNet-50 is frequently used as a backbone for satellite image classification, demonstrating strong results on both RGB and multispectral datasets.

Research on transfer learning confirms that fine-tuning ResNet-50 with full weight updates results in the highest cross-domain accuracy, with about 95.9% accuracy on unseen target domains after full fine-tuning, compared to around 88% with feature extraction alone. The residual architecture's capacity to preserve low-level spectral features while adapting higher-level semantic details makes it especially suitable for the cross-regional transfer task discussed here.

Domain shifts, characterized by statistical divergences between source (training) and target (testing) data, pose a known challenge in remote sensing. Studies on spatio-temporal transferability show that model performance drops significantly when applied across different regions and seasons without fine-tuning (Wang et al., 2025). Cross-continental transfer particularly suffers due to differences in spectral reflectance caused by atmospheric conditions, vegetation phenology, soil moisture, and building material variability.

A 2025 study in Nature Scientific Reports proved that self-attention mechanisms combined with optimized loss functions can greatly reduce domain shifts in satellite image classification tasks (Fang et al., 2025). Additionally, earlier work on cross-continental transfer (from Europe to Brazil) demonstrated that fine-tuning with small, region-specific samples can substantially improve classification accuracy, providing a theoretical basis for this study's experimental design. Bengisu et al. (2025) also showed that unsupervised domain adaptation can partially mitigate geographic differences without requiring labeled data from the target domain, although fine-tuning with labels remains more effective.

LULC studies in Lahore have mainly used traditional supervised classifiers Maximum Likelihood, Random Forest, and Support Vector Machines applied to Landsat-8 or Sentinel-2 imagery. A comprehensive 2024 study reported a 28.48% increase in settlement area from 1993 to 2023, with agricultural land decreasing from 62.9% to 23.54% of the district (Rahaman et al., 2024). Urban growth has averaged 12.1 to 26.4 km^2 per decade, driven by unplanned peri-urban expansion and densification (Yousaf et al., 2025).

These studies establish a baseline for LULC change in Lahore and highlight the landscape's heterogeneity and rapid transformation. However, no research has applied an EuroSAT-based deep learning framework to Lahore, nor assessed the transferability of EuroSAT-pretrained models to any South Asian city. Therefore, the gap in LULC transfer learning research in South Asia is both scientifically important and practically relevant consequential.

Identified Research Gap	How This Study Addresses It
EuroSAT evaluated only in European geographic contexts	Extends evaluation to Lahore (South Asian) as a novel target domain
No controlled study comparing ImageNet vs. EuroSAT pretraining	Directly compares both using a fixed ResNet-50 backbone
Zero-shot EuroSAT performance on South Asian imagery is unknown	Model A provides the first zero-shot EuroSAT evaluation on Lahore
Domain shift between Europe and South Asia has not been quantified	t-SNE and class-level ΔF1 analysis quantify the spectral and structural shift
Minimum fine-tuning sample size for South Asian LULC is unknown	Sample efficiency experiment (50–500 samples/class) with Model B
No labeled Sentinel-2 LULC dataset exists for Lahore	Study releases labeled dataset publicly under CC-BY 4.0 on Zenodo

6 Study Area

The study focuses on Lahore District in Punjab, Pakistan (31°15'N – 31°45'N; 74°01'E – 74°39'E). Covering about 1,772 km², it lies on the flat alluvial plains of the Upper Indus Basin and shares a border with India to the east. With an estimated population of over 14 million, Lahore is Pakistan's second-largest city and one of the fastest-growing urban areas in South Asia. The city experiences a semi-arid climate (Köppen BSh), featuring hot, dry summers (May–June, 40–45°C), a monsoon season (July–September), and mild winters. Its landscape is characterized by flat alluvial plains, irrigated agricultural fields (wheat, rice, mangoes), dense informal residential settlements, and a rapidly expanding peri-urban fringe. These features especially the spectral appearance of dry bare soil, irrigated crops, and mixed informal built-up areas significantly differ from the temperate, well-structured landscapes shown in the EuroSAT training data.

6.1 Justification for Study Area Selection

Lahore is selected as the study area for four key scientific reasons. First, it is among the fastest-urbanizing cities in South Asia, making it a critical and policy-relevant context where precise, scalable LULC monitoring is urgently needed. Second, Sentinel-2 images for the region are freely available via the

Copernicus Programme, with low cloud cover during the October–November post-monsoon season, ensuring consistent access to high-quality, cloud-free composites. Third, earlier LULC change studies conducted on Lahore using traditional classifiers offer a validated baseline for accuracy assessment and class boundary verification, allowing the deep learning results to be compared with established findings. Lastly, the researcher's position at Riphah International University and access to field sites within Lahore enable efficient, cost-effective ground truth data collection, reducing logistical challenges and the time required for field verification.

7 Research Methodology

7.1 Methodological Framework

This study uses a positivist, quantitative experimental method rooted in deep learning and remote sensing. The experiment primarily examines how the pretraining source (ImageNet versus EuroSAT) and the fine-tuning approach (zero-shot versus fine-tuned) influence LULC classification performance, with ResNet-50 serving as the consistent backbone for all scenarios. This controlled design ensures that any observed performance differences are directly attributable to the experimental variables, not architectural differences.

7.2 Phase 1

The EuroSAT Multispectral (MS) dataset will be sourced from Zenodo (DOI: 10.5281/zenodo.7711810). It contains 27,000 labeled image patches, each 64×64 pixels with a 10-meter resolution, spanning 13 Sentinel-2 bands and 10 LULC classes. The dataset is split into standard training (16,200), validation (5,400), and test (5,400) sets. Preprocessing includes normalizing all 13 bands per channel to a [1] range, based on the dataset-wide mean and standard deviation. During training, data augmentation will involve random horizontal and vertical flips, rotation up to ±15°, and brightness jitter of ±10%. No spatial resampling is needed since patches are already 64×64 pixels at 10 meters. A modified ResNet-50 backbone with a 13-channel input layer will be trained on EuroSAT, using the AdamW optimizer, cosine annealing learning rate schedule, and cross-entropy loss with label smoothing ($\varepsilon = 0.1$). The model with the highest validation accuracy will be saved as the best checkpoint, which will serve as the starting point for both Model A and Model B. Progression to subsequent phases requires a minimum overall test accuracy of 93% on EuroSAT.

7.3 Phase 2

Cloud-free Sentinel-2 Level-2A composite imagery for Lahore District (2023) will be accessed through Google Earth Engine (GEE), focusing on the October–November post-monsoon period to minimize cloud

cover and confusion from agricultural spectra. The composite will be generated using median pixel aggregation over a three-month period, with cloud masking from the QA60 band applied to each image prior to compositing. The imagery will be exported as GeoTIFF files at 10-meter resolution in EPSG:32643 (UTM Zone 43N), covering 10 spectral bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12). These will be divided into 64×64 pixel non-overlapping patches aligned on a regular grid over Lahore District, resulting in approximately 4,000–8,000 patches. Ground truth labels will be collected through a two-step procedure: first, visual interpretation of patches using high-resolution Google Earth Pro imagery in QGIS; second, field visits to clarify ambiguous or complex land cover classes. Each class will have at least 100 verified samples for training, with at least 30 samples per class for testing. The labeled dataset will be published on Zenodo under a Creative Commons Attribution 4.0 International (CC-BY 4.0) license.

7.4 Phase 3

Three experimental models will be trained and evaluated under tightly controlled, architecture-fixed conditions. The Baseline model uses a standard ResNet-50 with ImageNet pretrained weights from the torchvision model repository. Its final fully connected layer is replaced with a new linear layer that maps 2,048 features to 10 LULC classes. All layers are fine-tuned end-to-end on the Lahore training set using the AdamW optimizer with cosine learning rate decay over 30 epochs. Model A takes a different approach: a ResNet-50 with EuroSAT MS pretrained weights is applied directly to the Lahore test set without any weight updates or fine-tuning. This zero-shot setup tests the cross-regional transferability of EuroSAT features to South Asia, with class predictions mapped via a documented class-equivalence table. Model B combines both strategies: it initializes with EuroSAT MS pretrained weights and undergoes full-network fine-tuning on the Lahore training set, using the same hyperparameters as the Baseline AdamW optimizer, learning rate of 1×10^{-4}, cosine schedule, 30 epochs, batch size 16, and early stopping with patience of 5. This ensures a fair, directly comparable evaluation of all three models.

7.5 Phase 4

A sample efficiency test will only be conducted for Model B, involving fine-tuning with increasingly smaller training sets of 500, 300, 200, 100, and 50 labeled samples per class. Fixed random seeds (numpy.random.seed(42)) will guarantee consistent, nested subsampling. Each experiment starts from the EuroSAT checkpoint, and all five runs are tested on the same comprehensive Lahore test set. At each sample size, Overall Accuracy and Cohen's Kappa will be measured and plotted as a learning curve, aiming to find the smallest sample size that achieves an acceptable classification performance (OA > 85%).

7.6 Phase 5

Model performance will be evaluated using four complementary metrics: Overall Accuracy (OA), per-class F1-score, Cohen's Kappa (κ), and a comprehensive confusion matrix. Domain shift analysis will involve extracting 2,048-dimensional feature vectors from the penultimate layer of ResNet-50 (average pooling layer) for both EuroSAT and Lahore test images. These vectors will then be projected into two-dimensional space using t-SNE (perplexity = 30, 1,000 iterations, random_state = 42), allowing visualization of class cluster separation before and after fine-tuning. The per-class domain gap will be quantified as the difference in F1-score between EuroSAT performance and zero-shot Lahore performance for each class:

$$\Delta F1 = F1(\text{EuroSAT test, class}) - F1(\text{Lahore zero-shot, class})$$

A significant positive ΔF1 suggests weak transfer across regions for that class. Classes will be ordered by ΔF1 from highest to lowest, and spectral or structural reasons will be given for each class's rank ranking.

Table 1. Experimental Design Summary

Aspect	Baseline	Model A	Model B
Architecture	ResNet-50	ResNet-50	ResNet-50
Pretrained On	ImageNet	EuroSAT MS	EuroSAT MS
Lahore Fine-tuning	Yes	No (Zero-shot)	Yes
Purpose	Reference baseline	Measures raw transferability	Tests EuroSAT + adaptation
Lahore Training Data Used	Yes	No	Yes
Comparison Role	Lower bound	Mid reference	Expected best performer

The main hypothesis of the study is that Model B will achieve higher Overall Accuracy on the Lahore test set than the Baseline, with the Baseline surpassing Model A. Additionally, it is hypothesized that EuroSAT pretraining provides a benefit of at least 5–8 percentage points over pretraining only with ImageNet fine-tuning.

8 Expected Results and Contributions

Table 2. Anticipated Model Performance

Model	Expected OA (%)	Expected κ	Expected Macro F1
Model A EuroSAT → Zero-Shot	60–72%	0.55–0.68	0.58–0.70
Baseline ImageNet → Fine-tuned	78–85%	0.75–0.83	0.76–0.84
Model B EuroSAT → Fine-tuned	87–93%	0.85–0.91	0.86–0.92

Table 3. Expected Class-Level Transfer (Model A Zero-Shot)

LULC Class	Expected F1	Transfer Level	Primary Reason
Water Bodies / River	75–85%	High	Spectrally unique and globally consistent
Highway / Roads	65–75%	Moderate–High	Linear geometry universal across regions
Forest / Plantation	60–72%	Moderate	Spectral similarity but texture differences
Barren Land	50–65%	Moderate	Partially resembles EuroSAT Pasture class
Agricultural Crops	45–60%	Low	Irregular field patterns; mixed crop types
Residential Settlements	40–58%	Low	Dense informal housing vs. European housing

This study makes six original contributions to remote sensing and domain adaptation research. It presents the first controlled empirical evaluation of EuroSAT-to-Pakistan transfer learning with ResNet-50, using an experimental design that isolates the effects of pretraining source and fine-tuning strategy. Additionally, it offers a direct comparison between ImageNet and EuroSAT pretraining for Lahore LULC classification, maintaining a fixed ResNet-50 architecture to remove confounding variables. Beyond performance analysis, the study quantifies the domain shift between European and South Asian Sentinel-2 imagery by applying t-SNE visualisation and ΔF1 per class, revealing spectral and structural differences not previously assessed for South Asian cities. The research also provides the first publicly available labelled Lahore

Sentinel-2 LULC dataset (2023) under a CC-BY 4.0 license on Zenodo, with a permanent DOI, addressing the lack of open datasets for Pakistani remote sensing. Practically, it offers minimum sample size guidelines for South Asian LULC practitioners to achieve over 85% accuracy, based on a systematic experiment. All code, weights, and notebooks are published as a comprehensive, reproducible deep learning pipeline, suitable for other rapidly urbanising South Asian cities like Karachi, Islamabad, and Faisalabad.

9 Dataset and Software Requirements

Table 4. Datasets and its Sources

Dataset	Source	Format	Accessibility
EuroSAT Multispectral (13-band)	Zenodo DOI: 10.5281/zenodo.7711810	GeoTIFF	Free / Open Access
Sentinel-2 L2A (Lahore, 2023)	Google Earth Engine	GeoTIFF	Free / Open Access
Lahore District Boundary	GADM v4.1	Shapefile	Free / Open Access
Ground Truth Labels	Field + visual interpretation (QGIS)	CSV / Shapefile	Fieldwork cost only
High-resolution imagery (reference)	Google Earth Pro	Visual / KMZ	Free (academic use)

Table 5. Software Stack and its Purpose

Tool / Library	Version	Purpose
Google Earth Engine (JS API)	Latest	Sentinel-2 data acquisition and patch export
Python	3.10+	Core programming language
PyTorch	2.x	Deep learning model training and evaluation
torchvision	0.15+	ResNet-50 pretrained weights and image transforms

Rasterio	1.3+	Geospatial raster reading and patch extraction
GeoPandas	0.13+	Vector boundary data handling
Scikit-learn	1.3+	Confusion matrix, F1-score, Cohen's Kappa
Matplotlib / Seaborn	Latest	Results visualization and t-SNE plots
NumPy / Pandas	Latest	Array operations and data management
QGIS	3.x	Ground truth collection and study area mapping
ArcGIS Pro	3.x	Cartographic output and spatial analysis
Google Colab Pro / Kaggle		GPU compute environment (T4/A100) for model training
GitHub		Version control and reproducible research

10 Conclusion

This semester project explores the transferability of EuroSAT-pretrained ResNet-50 for LULC classification in Lahore, Pakistan, with focus on cross-regional application. It keeps the ResNet-50 backbone constant, varying only the pretraining source (ImageNet vs. EuroSAT) and fine-tuning method (zero-shot vs. fine-tuned). This approach offers a straightforward contribution to domain adaptation in remote sensing. The experimental design addresses practical questions for South Asian geospatial practitioners: Does EuroSAT pretraining outperform ImageNet? How much local data is needed for fine-tuning? Which land cover classes are most affected by regional domain shifts? Outcomes include a validated pipeline, a publicly available Lahore dataset, and minimum-sample guidance, benefiting academia and remote sensing professionals in Pakistan and South Asia. The researcher’s expertise in Sentinel-2 processing, deep learning, LULC classification, and geospatial analysis supports the project’s technical feasibility and timely completion within six months (April–October 2026).

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